

Telehealth vs. In-Person Care Outcomes Analysis

A Healthcare Data Science Case Study

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Project Overview

Objective

Evaluate telehealth versus in-person healthcare visits by analyzing patient outcomes, recovery time, satisfaction, treatment costs, health literacy, technology access, and lifestyle factors.

Dataset

Integrated 5 healthcare datasets

- Patient Clinical Data
- Employment & Insurance
- Device & Internet Access
- Health Literacy
- Lifestyle & Wellness

Tools

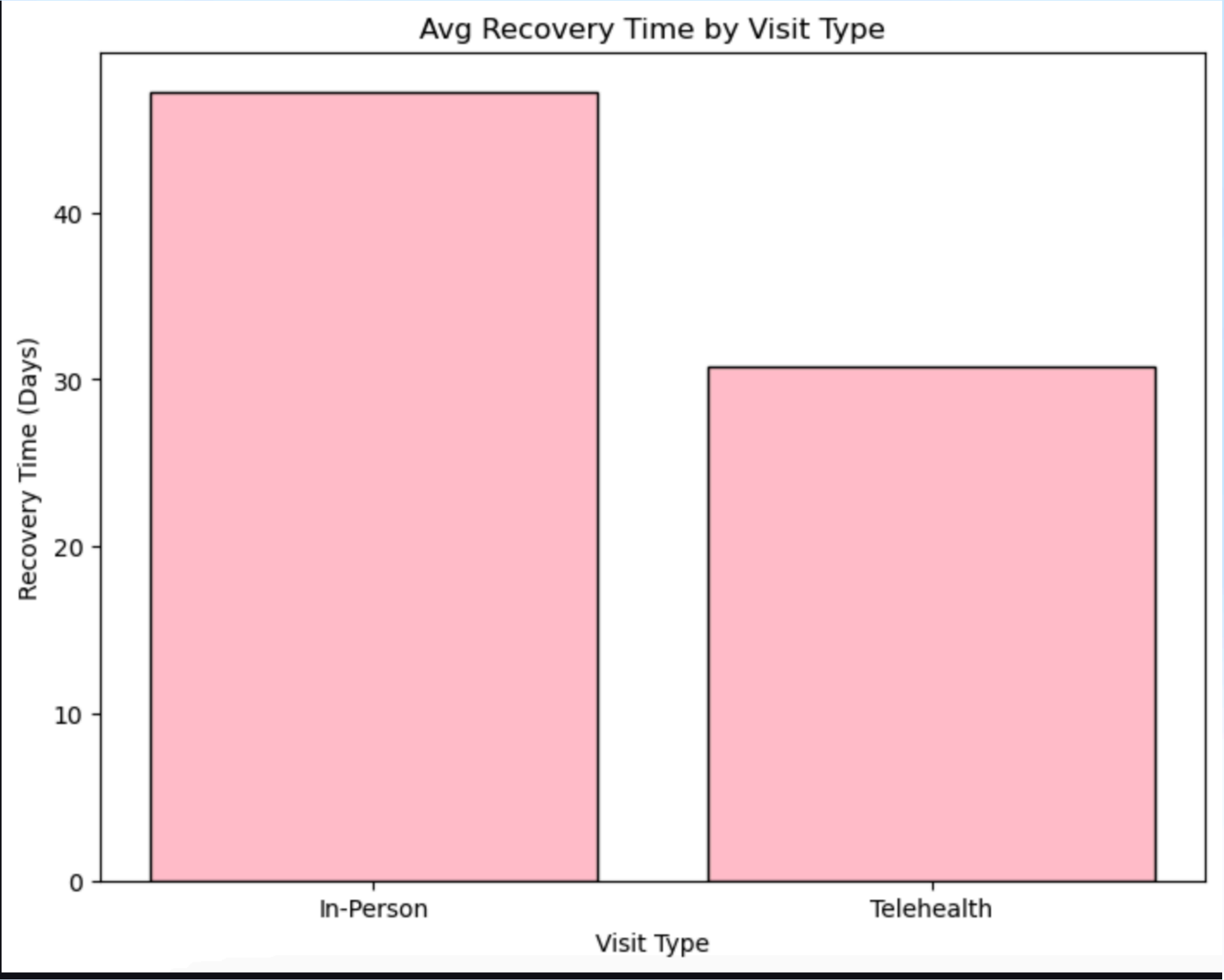
Python • Pandas • NumPy • Matplotlib • Jupyter Notebook

Workflow

■ Import Data → ■ Clean Data → ■ Merge Data → ■ Feature Engineering → ■ Statistical Analysis → ■ Business Insights

Key Findings

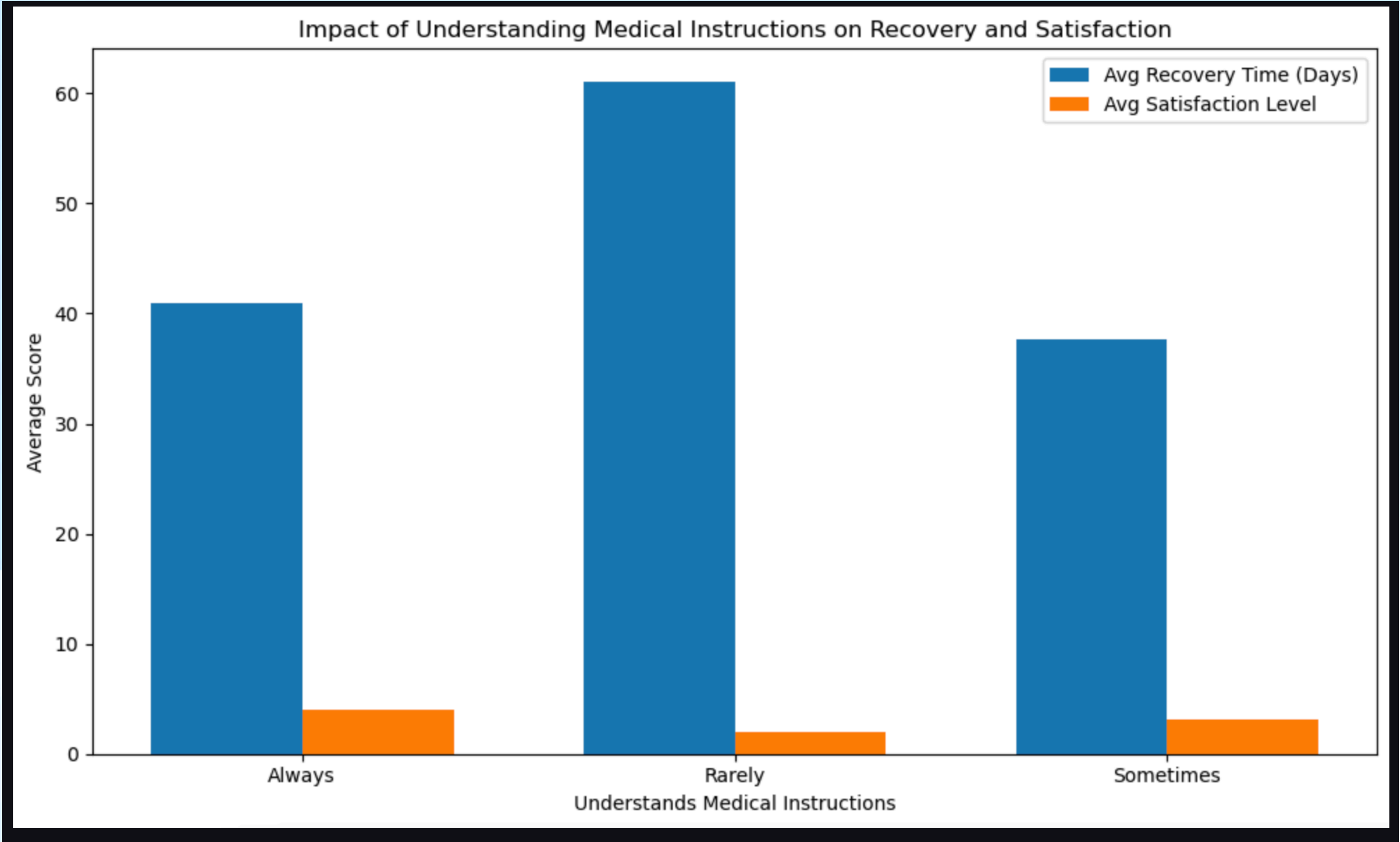
Telehealth Reduced Average Recovery Time



Key Insight

Patients receiving telehealth services recovered approximately 16 days faster than those receiving in-person care, highlighting the potential of telehealth to improve care efficiency while maintaining positive patient outcomes.

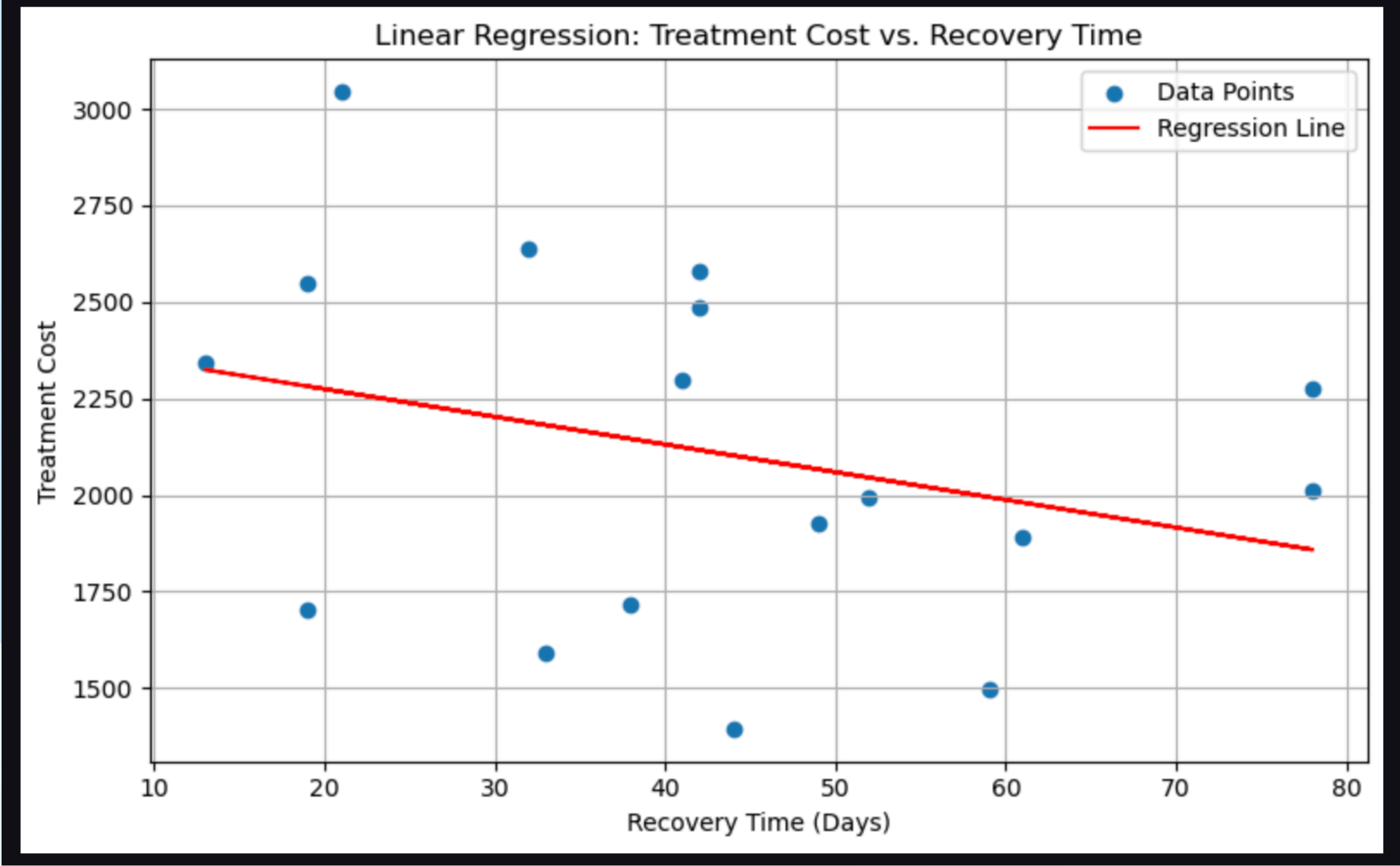
Health Literacy & Patient Outcomes



Key Insight

Patients who consistently understood medical instructions experienced shorter recovery times and higher satisfaction, highlighting the importance of health literacy in care delivery.

Statistical Analysis



Key Insight

Regression analysis explored the relationship between treatment cost and recovery time. The results indicate that higher treatment costs did not consistently lead to faster recovery, emphasizing that patient outcomes depend on multiple clinical and behavioral factors.

Project Impact

Project Highlights

- Evaluated the effectiveness of telehealth compared with traditional in-person care using multiple healthcare datasets.
- Identified factors influencing patient recovery, satisfaction, and treatment costs.
- Generated data-driven insights that can support healthcare organizations in improving care delivery and patient outcomes.

Delivered actionable healthcare insights

- ✓ Integrated five healthcare datasets
- ✓ Performed statistical and exploratory data analysis
- ✓ Visualized patient outcome trends
- ✓ Developed actionable healthcare insights

Final Statement

This project demonstrates an end-to-end healthcare analytics workflow, from integrating multiple datasets and performing statistical analysis to delivering business insights that support evidence-based healthcare decision-making.